MSDS 570 Visualization and Unstructured Data Analysis

Homework Assignment 1: Introduction to Different Type of Plots

Due Monday, Feb 26th, 2023 (11.59 pm CST)

Total 100 points

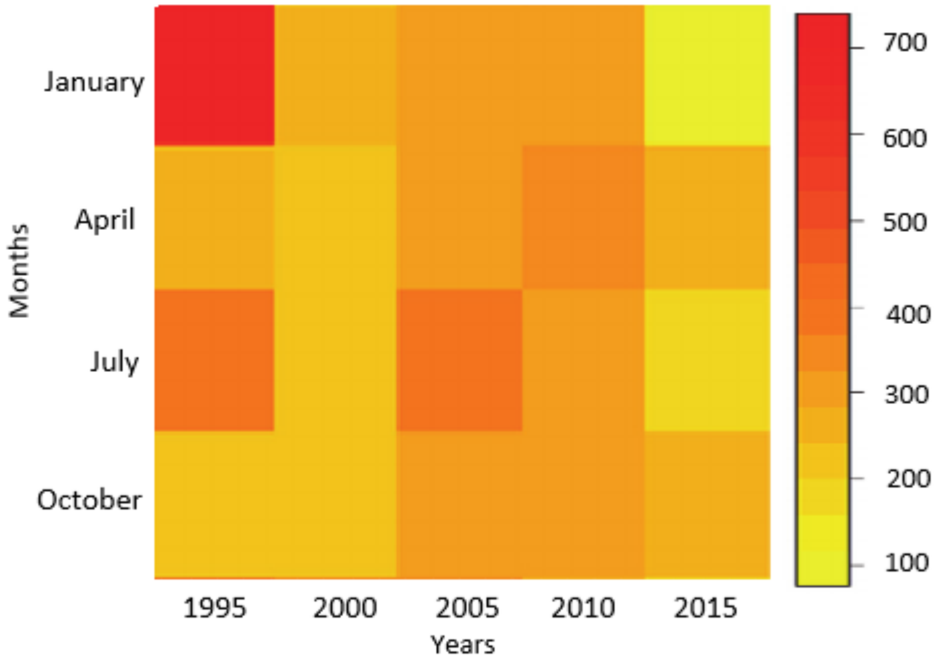
Homework Submission instructions

To submit this homework:

- Please create a new notebook named as **Name_MSDS570_HW#1**.
- Please write your solutions and submit the notebook files on Blackboard or email me if facing issues in submitting the notebook file.
- Assignments **MUST** be submitted on time.
- This programming assignment **should be done individually**. You can discuss the problems but the codes should not be same.
- Late submission policy will be applied for the late submissions
- Feedbacks will be provided in your solution notebook below the questions

Problem 1 (25 Points)

You are given below diagram that gives information about the road accidents that have occurred over the past two decades during the months of January, April, July, and October:

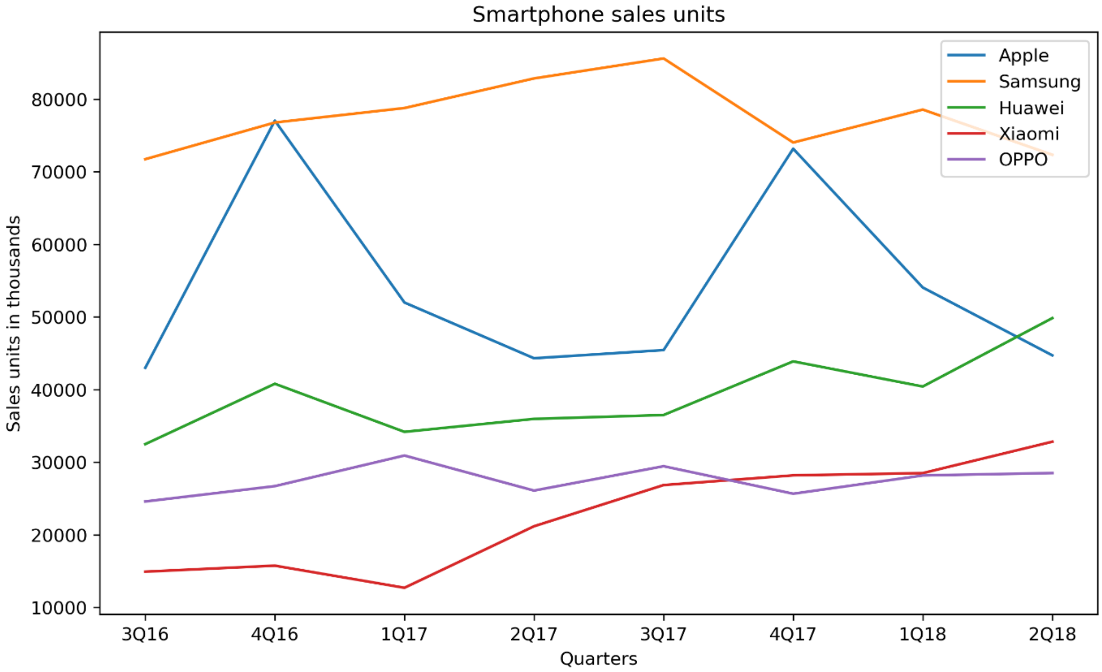


(a) Identify the year during which the number of road accidents occurred were the least.

(b) For the past two decades, identify the month for which accidents show a marked decrease:

Problem 2 (25 Points)

You want to compare smartphone sales units for the five biggest smartphone manufacturers over time and see whether there is any trend:

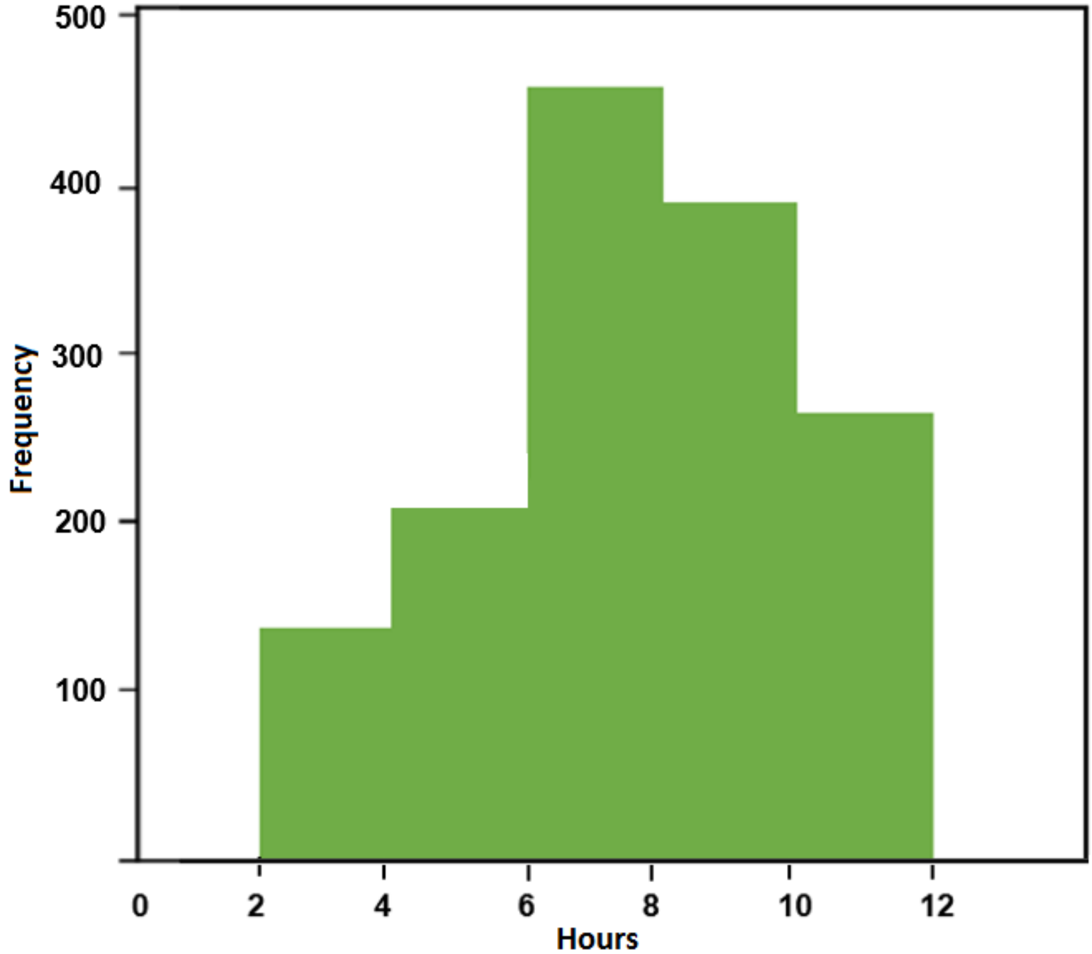


(a) Looking at the above line chart, analyze the sales of each manufacturer and identify the one whose performance is exceptional in the fourth quarter as compared to the third quarter.

(b) Analyze the performance of all manufacturers, and make a prediction about two companies whose sales unit will show a downward trend and an upward trend.

Problem 3 (25 Points)

You are provided with a histogram that states the total number of trains arriving at different time intervals:



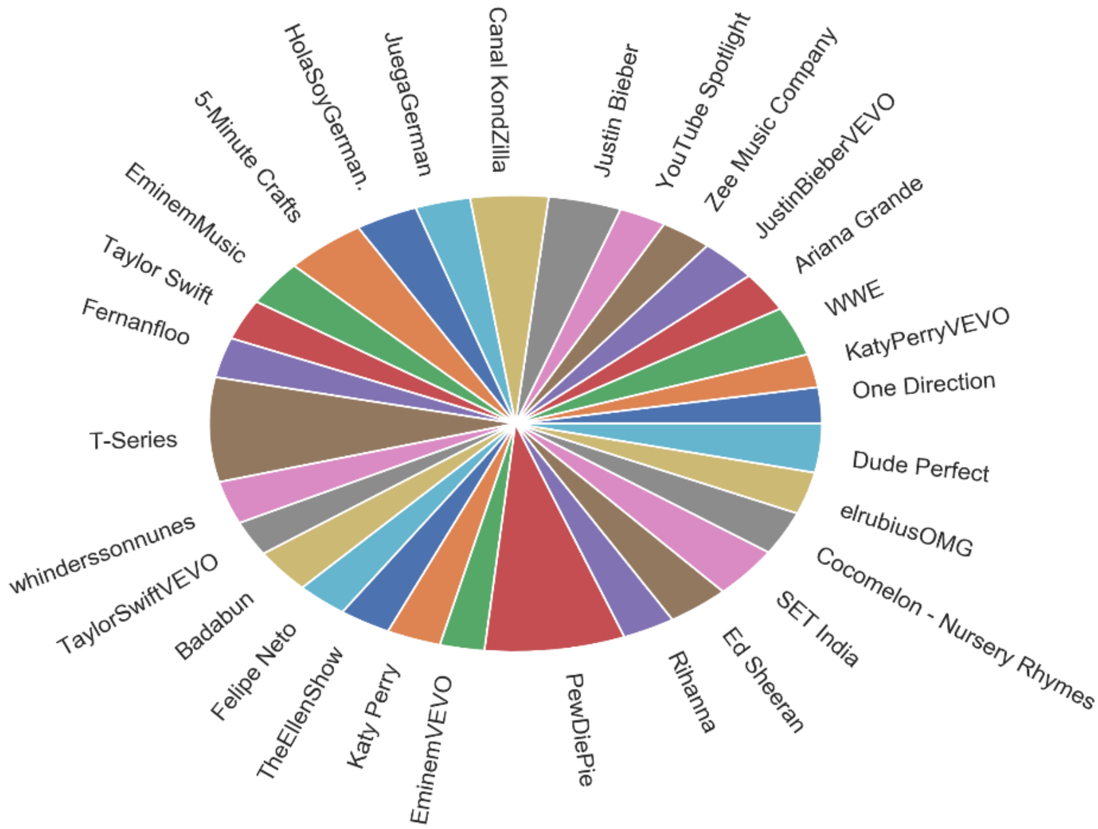
(a) By looking at the above graph, can you identify the interval during which the most number of trains arrive?

(b) How would the histogram change if the number of trains arriving between 4 and 6 pm were to be increased by 50?

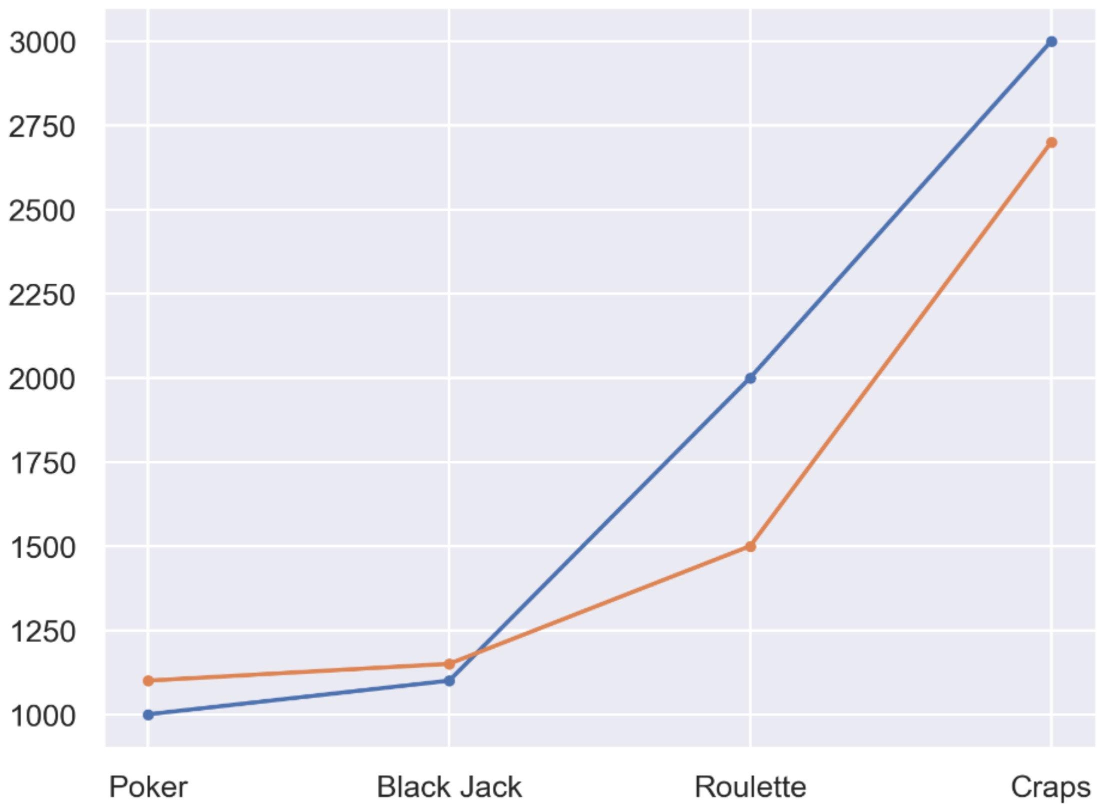
Problem 4 (25 Points)

The following visualizations are not ideal as they do not represent data well. Answer the following questions for each visualization:

The first visualization is supposed to illustrate the top 30 YouTubers according to the number of subscribers:



The second visualization is supposed to illustrate the number of people playing a certain game in a casino over two days:



(a) What are the bad aspects of these visualizations?

(b) How could we improve the visualizations? Propose the right visualization for both scenarios.